

Liam Willis

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Education

- University of Massachusetts Boston | Bachelor of Science in Computer Science | 4.0 GPA | Expected December 2025
- Sturgis Charter Public School | High School Diploma | International Baccalaureate Diploma | 4.18 GPA | Graduated June 2021

Skills

- I am proficient with the following languages and technologies: HTML, CSS, JavaScript (three.js), Java, Python, C, SQL, Assembly Language, Coq, WebGL/GPU, Git/GitHub, Excel, and PowerShell.

Relevant Coursework

- Computing in Python
- Intermediate Computing with Data Structures
- Programming in C
- Calculus-Based Probability and Statistics
- Operating Systems
- Computer Graphics
- Database Design
- Discrete Mathematics
- Linear Algebra
- Advanced Algorithms
- Theory of Computation
- Compilers

Projects

DATA COMPRESSION ALGORITHM

- Implemented the Huffman Encoding Algorithm in C, compressing text files into reduced-sized binary files
- Utilizes a minimum heap and a binary tree to construct binary codes for each data element, and employs bitwise operations to write the codes to the compressed binary file

KD-TREE IMPLEMENTATION

- Implemented a symbol table API in Java whose keys are two-dimensional points using both a red-black binary search tree and a 2D-tree as underlying data structures.
- Supports search, insert, traversal, nearest-neighbor search, and k-nearest neighbor search operations.

ERROR-CORRECTING CODE ALGORITHM

- Implemented the Hamming (7, 4) error-correcting code algorithm in C, capable of correcting incorrect data bits which may occur during data transmission.
- Utilizes bitwise operations to calculate parity of data bits, store them in seven different disks, and retrieve them later to determine potential flipped bits

Accolades

- Recipient of the Rensselaer Medal Award for outstanding achievement in Mathematics and Physics.
- Recipient of the 2021 Harbor One Community Scholarship for community service and academic achievement.
- Recipient of the UMass Amherst Cape Cod Alumni Network Scholarship for excellent academic performance.